

CYBORG NEWS

CYBORGNEWS.MIT.EDU

THURSDAY, SEPTEMBER 10, 2026 · VOL. II · NO. 9

FREE TO THE LAB

F'MATH · ASTRA
- 97.6 unchangedF'MATH · FABLE
▲ 87.8 +9.8 revisedEXPLOITBENCH
▲ 11.5 +6.0 revisedHALLUC · ASTRA
- 4.2% revertedLEAN THMS · FLT
▲ 29,500 11 daysCACHE READ/MTOK
▼ \$0.25 -50%INJECTION BOUNTY
▲ \$130K per userWhat's
News

Research & Models

❏ **A machine-written proof** of Fermat's Last Theorem cleared the Lean kernel after eleven days of largely autonomous work, at 13 million lines and 29,500 lemmas.

❏ **OpenAI edited several** of the benchmark figures in its Astra announcement after publishing it, including a rival lab's math score, and reverted one of them.

❏ **Meta says it designed** its Muse agent on the assumption that the agent itself will be compromised, isolating each user in a dedicated Linux virtual machine.

❏ **Data work, not model work**, accounted for the larger share of compute-efficiency gains between 2019 and 2025, a 3.24x factor that reframes the scaling argument.

❏ **World Labs released Atlas**, an omni world model that treats text, image, video and 3D as one autoregressive diffusion sequence.

Around the Institute

❏ **A new method called CW-Net** translates the internal state of a self-driving stack into concepts a person can argue with, rather than a saliency map.

❏ **A protein-design framework** from campus raises the success rate of computational design while deliberately steering away from sequences already found in nature.

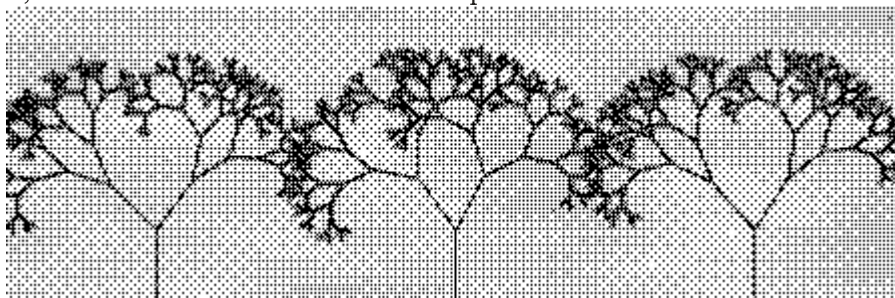
❏ **The QMIT fellows arrive** this term; the lab hosts two of the cohort and will share the Building 34 cryostat through the spring.

❏ **Reading group resumes** Tuesdays at 4, with the formalization papers first on the list. Sign-up sheet is on the door.

FORMAL METHODS

The Proof That
No Human Wrote

Claude closed Fermat's Last Theorem in Lean in eleven days — 13 million lines, 29,500 lemmas, and no human in the verification loop.



DEPENDENCY The formalization proceeded as a branching dependency graph: every lemma had to be discharged before the statements above it could close.

CYBORG NEWS ILLUSTRATION

CAMBRIDGE, MASS. — A language model has produced the first complete, machine-checked proof of Fermat's Last Theorem, finishing in eleven days a formalization that had occupied a well-organized human project for years.

The proof is written in Lean and runs to roughly 13 million lines, establishing some 29,500 intermediate theorems along the way —

about five times the bulk of Mathlib, the library the Lean community has been assembling for the better part of a decade. Kevin Buzzard, the Imperial College London mathematician who has led the human effort to formalize the theorem, reviewed the output. It “proves Fermat's Last Theorem with no assumptions other than the axioms of mathematics,” he wrote.

Machines have been checking proofs since the 1970s. What is new is that a machine wrote the object being checked, at a scale no reading group could audit by hand, and a far simpler program then confirmed it without anyone having to trust the first. The Lean kernel that validated 13 million lines is a few thousand lines long and has been read by humans.

BENCHMARKS

The Scores Moved
After the Post Went Up

OpenAI revised several figures in its Astra announcement — including a competitor's — then put one of them back.

OpenAI's Sept. 3 announcement of GPT-6 Astra did not load properly, was pulled shortly after 2 p.m., and returned with several of its evaluation numbers changed.

The reported hallucination rate for Astra fell from 4.2 percent to 2 percent, and then went back to 4.2. The cybersecurity score for GPT-5.6 Sol on ExploitBench

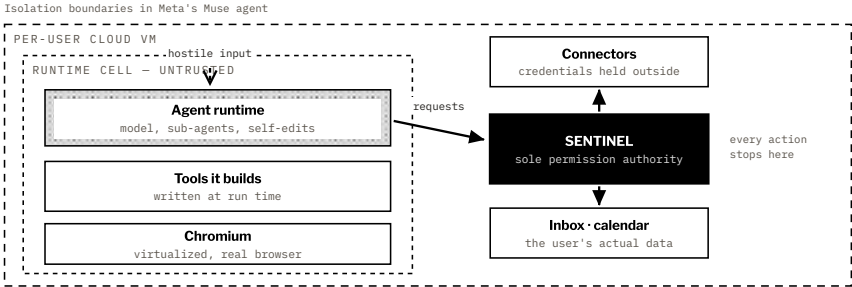
roughly doubled, from 5.5 to 11.5. The coding number moved from 57.7 to 57.9. The change that drew the most attention was not to OpenAI's own model: Anthropic's Fable 5.1 appeared on FrontierMath first at 78 percent and later at 87.8 — a rival's score, revised upward, on a chart built to compare it against Astra's 97.6.

“We care deeply about getting evaluations right,” a spokesperson said. “Most evaluations have noise within a few percentage points.” That is true, and it is also true that noise is usually symmetric. “It's usually a function of final launch logistics,” said Vincent Sunn Chen of Snorkel AI.

AGENT SAFETY

Meta Built Muse Expecting It to Be Attacked

Every user gets a Linux virtual machine, the agent runtime is treated as hostile, and a separate service holds the only key to the outside world.



HOW IT IS FENCED The agent runtime is given no credentials of its own. Anything that touches the outside world is a request to Sentinel, which runs outside the cell and can refuse.

CYBORG NEWS · AFTER META SUPERINTELLIGENCE LABS

MENLO PARK, CALIF. — Meta published the security architecture behind Muse, its personal agent, and the notable thing about the document is what it declines to promise.

Muse reads your inbox, holds your calendar, writes its own tools, runs a real browser and can edit itself. Meta's answer to the obvious objection is not that the agent

will behave. It is that the agent is assumed not to.

“We designed the system to assume the agent may be under attack, and to limit the potential damage.”

TAREK SHEASHA, META
SUPERINTELLIGENCE LABS

Each user gets an isolated Linux virtual machine — a dedicated cloud computer rather than a tenant in a shared one. Inside it, Linux isolation separates the agent runtime from the security services, so the part of the system that reads untrusted text is not the part that holds credentials. A service called Sentinel is the sole permission authority for every external action: the runtime

THE WEEK AHEAD

- TUE 16:00 **Reading group: FLT in Lean**
34-401 · bring the blueprint
- WED 11:00 **Agent sandboxing review**
Safety working group
- WED 15:30 **Cluster maintenance**
No jobs after 15:00
- THU 12:00 **Visitor: World Labs**
Atlas walkthrough
- FRI 10:00 **Eval harness freeze**
Seeds and configs to the repo

FERMAT, BY THE NUMBERS

11 days
WALL-CLOCK TIME, LARGELY AUTONOMOUS

13 million
LINES OF LEAN PRODUCED

29,500
INTERMEDIATE THEOREMS DISCHARGED

6 billion
OUTPUT TOKENS CONSUMED

Anthropic, Sept. 4, 2026.

SCALING

The Gains Came From the Data

Between 2019 and 2025, improvements to training data delivered 3.24 times more compute-efficiency than improvements to models.

3.24×
MORE COMPUTE-EFFICIENCY FROM DATA WORK THAN FROM MODEL WORK

2019–25
PERIOD THE DECOMPOSITION COVERS

1 line
WHERE THAT WORK USUALLY APPEARS IN A PAPER

A decomposition of where compute-efficiency gains actually came from between 2019 and 2025 puts 3.24 times more of the improvement on data work than on model work.

This is not the story the field tells about itself. The public account of the last six years is architectural: atten-

tion variants, normalization tricks, optimizer schedules, mixtures of experts. Data work is the part that appears in a footnote as “we filtered the corpus.”

The finding suggests the footnote was doing most of the lifting — deduplication, quality filtering, mixture weighting, synthetic aug-

mentation and the unglamorous business of deciding what a model should not read.

Two consequences follow for a small lab. Data work is comparatively cheap and does not need a frontier cluster.

EDITED AFTER PUBLICATION

ASTRA ANNOUNCEMENT, SEPT. 3

BENCHMARK	FIRST	THEN	Δ
ExploitBench · Sol	5.5	11.5	+6.6 ▲
FrontierMath · Fable 5.1	78.0	87.8	+9.8 ▲
Coding · Astra	57.7	57.9	+0.2 ▲
Hallucination · Astra	4.2	2.0	-2.2 ▼
Hallucination · reverted	2.0	4.2	+2.2 ▲

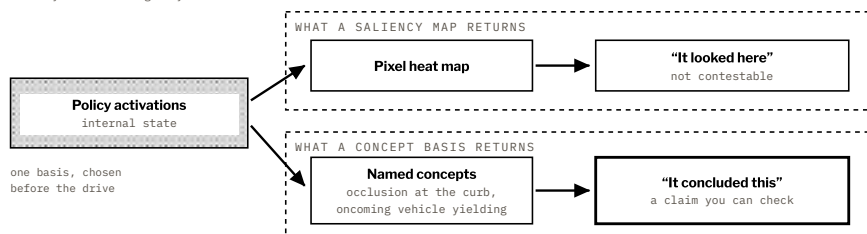
Fortune, Sept. 4. Lower is better for hallucination rate.

INTERPRETABILITY

Teaching a Driving Stack to Explain Itself

CW-Net converts the internal state of a driving policy into named concepts, so a disagreement about a decision can be an argument rather than a heat map.

Two ways of answering "why did it brake?"



CONCEPT BASIS A heat map reports where the network attended. A concept basis reports what it concluded, in terms fixed before the drive — which is the form a claim has to take to be checked.

CYBORG NEWS

CAMBRIDGE, MASS. —A method developed on campus, CW-Net, translates the reasoning of an autonomous vehicle's perception-and-control stack into concepts a person can name, and therefore contest.

The prevailing tool for this has been the saliency map, which shades the pixels a network attended to. Saliency maps are popular because

they are easy to produce and easy to look at, and they are close to useless in a review meeting: a bright patch on a pedestrian answers a question nobody asked. It tells you where the network looked, not what it concluded.

CW-Net instead projects internal activations onto a basis of concepts with names — occlusion at the curb, oncoming vehicle yielding,

lane boundary ambiguous — and reports which of them carried a given decision, and with what weight.

The practical difference shows up in failure analysis. An engineer can state a hypothesis about why the car braked, check it against the concept weights, and be wrong in a way that is visible to everyone in the room.

HOUSE RULE OF THE WEEK

Publish the run you did first. If a number moves between draft and talk, the new one goes on the slide and the old one in the appendix, with date and seed.

AROUND THE INSTITUTE

QMIT Fellows Arrive

The first cohort of QMIT Fellows joins this term. We host two of them, and the Building 34 cryostat is shared through the spring — book it early, and read the handling note before you touch the transfer line.

OVERHEARD

"It is not autonomous if you are the one restarting it."

EMBODIMENT

Whole-Body Reasoning Leaves the Lab

Robotic systems are converging on whole-body reasoning — planning that treats locomotion, manipulation and perception as one problem rather than three pipelines with a message bus between them.

The application list is unchanged from a decade ago: warehouses, laboratories, farms, care settings, factory floors. What has changed is that the demonstrations are now happening in rooms the researchers did not build.

That shift is easy to undersell. A policy that works in a room it has seen is a controller; a policy that works in a room it has not is the beginning of a product. The gap between them has been the whole field for twenty years,

and it is finally being crossed in public, on video, by more than one group.

The lab's own position is that the bottleneck has moved from control to specification. We can make the arm do the thing. Writing down what the thing is, in a form that survives contact with an unfamiliar kitchen, remains open — and it is a language problem at least as much as a dynamics one.

Which is why the world-model work and the manipulation work keep converging on the same question. A planner needs to predict the next state of a scene in geometry, not in pixels, and it needs to be told what counts as success in terms a person would recognize.

MACHINE LEARNING FOR DESIGN

Designing Proteins Nature Never Wrote

The success rate goes up, and the sequences deliberately move away from anything already in the database.

A machine-learning framework for computational protein design, developed at the Institute, improves the rate at which designed sequences fold as intended while explicitly penalizing similarity to natural sequences.

The second half is the interesting half. A generative model trained on the protein data bank has an easy way to score well: return something close to a sequence it has already seen. The design is plausible, the metrics look excellent, and nothing has been discovered.

By pushing away from the training distribution while holding the structural objec-

tive, the framework is forced to find genuinely novel solutions to the same folding problem — which is what design, as opposed to retrieval, means.

The same failure mode is worth naming outside biology. Any generative system evaluated on its resemblance to a reference corpus will be rewarded for memorizing it, and most benchmarks in this building are still scored that way.

REMINDER

Reading group is Tuesday at four. 34-401.